



Progressive Education Society's
Modern College of Engineering, Shivajinagar, Pune-05.
Department of Electronics and Telecommunication Engineering
Academic Year- 2025-26

Project ID: AE-2026-ET14

Project Title: Real time water pipeline leakage detection using IoT

Name of the Students: 1. Rushikesh Tanaji Patil 2. Ashitosh Rangnath Jadhav 3. Sakshi Raju Baravkar

Objectives: 1 To monitor pressure and flow rate using sensors
2 To detect water leakage in pipelines in real-time.

Methodology:

- Hardware Setup:**
ESP32/Arduino with flow and pressure sensors, LoRa module for communication. Solenoid valve with relay, LCD/LED/buzzer, and battery with solar
- Firmware Development :**
Microcontroller reads flow and pressure sensor data and processes it continuously. Processed data is transmitted to the cloud and used to detect leakage conditions.
- Cloud Integration :**
Sensor data is transmitted to the cloud platform via Wi-Fi/LoRa for storage and processing. Real-time monitoring is enabled through a mobile or web dashboard with instant updates.
- Dashboard & Alerts :**
Real-time data is displayed on a mobile/web dashboard for continuous monitoring. Alerts are generated through buzzer and IoT notifications when leakage is detected.

Block Diagram:

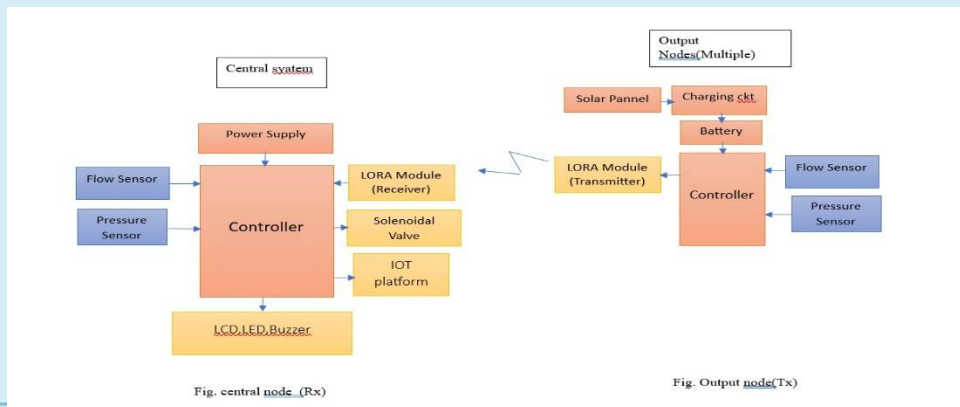


Fig. central node_(Rx)

Fig. Output node(Tx)

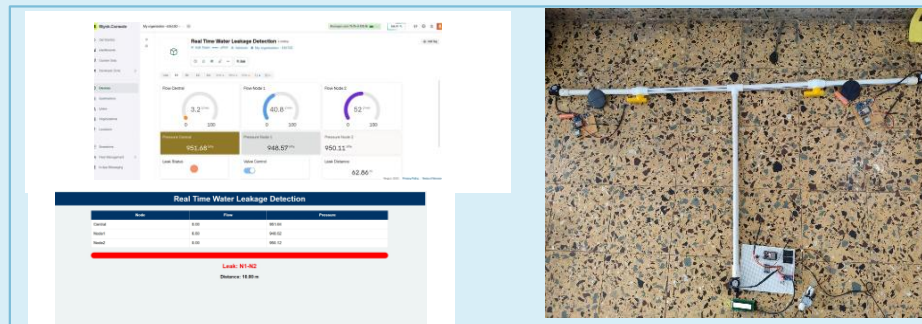
Testing and Debugging:

- Sensor Calibration:** Flow and pressure sensors validated under normal and leakage conditions to ensure accurate measurement of water parameters.
- Connectivity:** ESP32 data transmission via LoRa/Wi-Fi tested continuously; system performance verified during network interruptions.
- System Reliability Testing:** The system was tested for long-duration operation under varying conditions.
- Dashboard:** IoT dashboard tested for real-time monitoring; alert system verified for buzzer and

Specifications:

- Hardware:** ESP32/Arduino, Flow Sensor, Pressure Sensor, LoRa (Solenoid Valve, Relay Module, LCD, LED, Buzzer, Battery/Solar
- Software:** Arduino IDE (Embedded C firmware), IoT Dashboard (Blynk/ThingSpeak), optional backend for data monitoring.
- Leak Detection:** Based on comparison of flow and pressure values; detects abnormal variations for real-time
- Connectivity:** Wi-Fi + LoRa communication; supports MQTT/HTTP protocol for cloud integration.
- Power:** 55V/12V adapter or battery with solar panel and charging Circuit.

Result (Photo of Working Project):



Conclusion: The system successfully integrates IoT sensors, ESP32/Arduino processing, and a cloud platform to deliver real-time water pipeline leakage detection with continuous monitoring of flow and pressure parameters and instant alert generation. The system is cost-effective, energy-efficient, and scalable for households, industries, and smart city water management — addressing the critical issue of water wastage. Future scope includes AI-based leakage prediction, LoRa-based wide area monitoring, and automated valve control for improved efficiency.